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Eclampsia and preeclampsia with tropical seasonality: Evidence from Indonesia's national health insurance registry (2015–2022)

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ABSTRACT

Background: Climate exhibits substantial variation based on geographic location and latitude. Evidence of its effects on maternal health largely comes from temperate regions, while studies in tropical areas remain scarce, necessitating region-specific research to address these gaps effectively. This study assessed the association between climatic factors, such as temperature, precipitation, humidity, and solar radiation, and eclampsia and preeclampsia.

Methods: This study used health insurance data from Indonesia's National Health Insurance registry, covering 342,432 pregnancies across 514 districts in 34 provinces from 2015 to 2022. Associations between climatic factors and pregnancy complications were analyzed using multilevel Poisson regression models, adjusting for individual and district characteristics. The relative risk (RR) and 95% confidence intervals (CI) were used to assess these relationships.

Results: A 5 °C increase in temperature was associated with a decreased risk of pregnancy complications (RR = 0.86, 95% CI: 0.83–0.89), and a 100 mm increase in precipitation also lowered the risk (RR = 0.97, 95% CI: 0.95–0.98). Conversely, higher solar radiation (per 50 W/m^2) and humidity (per 10 g/m^3) were linked to an increased risk, with RRs of 1.02 (95% CI: 1.01–1.04) and 1.12 (95% CI: 1.05–1.19), respectively. Adjusting for district characteristics did not alter these relationships significantly.

Conclusion: This study highlights the impact of climatic factors on the incidence of eclampsia and preeclampsia. A coordinated effort among multi stakeholders is crucial, with policymakers prioritizing the inclusion of localized climate variables in health programs, ensuring effective policy implementation, and preparing healthcare providers for climate-related challenges.

1. Introduction

The climate change effects on maternal health present substantial challenges to maternal well-being on a global scale. In 2020, the World Health Organization (WHO) estimated that maternal mortality rate (MMR) in low- and middle-income countries were higher than in high-income countries, with 430 compared to 12 per 100,000 live births [1]. Most maternal deaths are mostly due to blood loss and hypertension [2] which are attributable to eclampsia and preeclampsia. Climate change compounds these risks by increasing threats to maternal health [3] as the human body is particularly sensitive to environmental changes, adapting through homeostatic mechanisms involving the autonomic nervous, endocrine, and cardiovascular systems [4–6]. For example, vasoconstriction adjusts in response to temperature changes

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2210-7789/© 2026 International Society for the Study of Hypertension in Pregnancy. Published by Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

[7] and the body's ability to cool itself through sweating is influenced by the surrounding humidity levels [8]. Moreover, rising temperature, floods, heatwaves, and drought can inflict damage on healthcare infrastructure as well [9–11], particularly in low-resource settings, hindering pregnant women's access to essential maternal care and induce psychosocial stress, heightening complications risks during pregnancy [12,13].

Previous empirical studies investigating seasonal and climatic factor impacts on eclampsia and preeclampsia have reported mixed and often inconsistent findings, highlighting the complexity of these interactions and raising questions about the generalizability of findings across diverse geographical settings. A study in India observed that periods with lower temperatures and elevated humidity levels were associated with an increased incidence of eclampsia [14]. In contrast, heatwaves in United States, characterised by higher temperature, elevated the risks [15]. A study in South Africa reported no impact of precipitation on the incidence of eclampsia or preeclampsia, while research in Thailand identified an elevated risk of these conditions during periods of low precipitation [16,17]. In Zimbabwe, elevated humidity levels have been associated with a significant increase in the incidence of eclampsia [18]. Algert et al. (2010) identified a correlation between increased sunlight and a reduction in the incidence of preeclampsia in Australia [19]. A systematic review by Hlimi (2015) highlights the diverse impacts of climatic factors on eclampsia and preeclampsia among Sub-Saharan African and Central and South Asian countries, indicating notable differences between temperate and tropical regions [20].

Maternal mortality in Indonesia remains a significant concern. Although the MMR decreased from 346 deaths in 2010 to 305 deaths per 100,000 live births in 2015 [21], these numbers fail to meet The Indonesia's National Medium-Term Development Planning (*Rencana Pembangunan Jangka Menengah Nasional* – RPJMN) target of 183 per 100,000 live births by 2024 [22]. In 2015, preeclampsia accounted for about 24% of maternal deaths in Indonesia, with an annual incidence of eclampsia and preeclampsia estimated at 3.8% to 8.5%, surpassing the regional Asian averages of 0.2% to 6.7% [23–25]. Despite the Indonesian government's efforts to address maternal health issues and its commitment to reducing carbon emissions [26], climatological factors have not yet been incorporated into maternal health strategies. This omission may lead to the development of ineffective measures for preventing maternal health problems.

Indonesia is an archipelago with diverse climatic condition across different regions and districts. Moreover, decentralised governance in Indonesia has highly contributed to different local maternal health strategies and climate-related policy at sub-national settings. Despite existing literature gaps, a comprehensive assessment of climate change indicators can inform targeted strategies to mitigate maternal health burdens. Therefore, this study aims to assess the impact of climatic factors such as temperature, solar radiation, precipitation, and humidity on pregnancy complications incidence at the sub-national level in Indonesia.

2. Methods

2.1. Data source, population and sampling

This study used *Jaminan Kesehatan Nasional* (JKN)—national health insurance—obtained from the *Badan Penyelenggara Jaminan Sosial Kesehatan* (BPJS)—Indonesia's national Social Security Agency on Health. The registry stores records of individuals enrolled in JKN, including healthcare visits, referrals, medical procedures, diagnoses, prescribed medications, treatment durations, and associated expenses. By January 31, 2023, JKN had enrolled around 249.67 million individuals, accounting for roughly 90% of Indonesia's whole population [27]. We employed the data sample registry between 2015 and 2022, encompassing 1,648,135 medical treatment records derived from 324,819 women who were pregnant and were in breastfeeding period. Further details about the JKN data sampling process can be found elsewhere [28]. We identified pregnant individuals, resulting in a dataset of 479,622 medical visits from 292,053 individuals. From this final cohort, there were 342,432 pregnancies distributed across 514 districts in 34 provinces.

The climate dataset was compiled from two primary sources, i.e. ERA5 and observation data from ground-based station owned by the Indonesia's Climatology, Meteorological, Climatological, and Geophysical Agency (Badan Meteorologi, Klimatologi, dan Geofisika or BMKG). ERA5 is a global climate reanalysis model developed by the European Centre for Medium-Range Weather Forecasts. The dataset is available on an hourly time scale with a spatial resolution of about 25 km by 25 km. From ERA5, we selected several climate parameters, including 2-m air temperature (°C), dew-point temperature (°C), and downward shortwave solar radiation (W/m^2). Relative humidity (%) is estimated using an empirical approximation based on air temperature and dew-point temperature, as it cannot be measured directly. Precipitation (mm) is provided by BMKG through statistical downscaling, which incorporates data from reanalysis models, ground-based observations, and satellite images. The precipitation parameter is provided as a monthly accumulation value.

2.2. Outcomes

This study identified pregnancy complications from two major conditions frequently associated with maternal mortality: preeclampsia and eclampsia [29]. The International Classification of Diseases, 10th Revision (ICD-10), was utilised to classify these complications within the dataset. Specifically, the ICD-10 codes O14 and O15 correspond to preeclampsia and eclampsia, respectively. The proportion of pregnancy complication among all pregnancies was calculated monthly from January 2015 to December 2022 and for all districts.

2.3. Exposure

The exposure variable is defined by the monthly averages of key climatic factors measured within districts from January 2015 to December 2022, namely temperature, precipitation, relative humidity, and solar radiation at surface. Temperature, measured in degrees Celsius, provides insight into the thermal conditions of a particular area. Precipitation, measured in millimetres, reflects the amount of accumulative rainfall in a certain period. Relative humidity, expressed as a percentage, signifies the water vapor content in the air, influencing atmospheric conditions. Solar radiation at surface, measured Watt per meter square (W/m^2), represents the intensity of sunlight exposure at the horizontal surface.

Individual characteristics included in this study consisted of gender (male or female), age (in years), and membership type. Membership in JKN system is divided into two groups: PBI (*penerima bantuan iuran* or subsidy beneficiaries) and non-PBI (non-subsidy beneficiaries). PBI members are identified as individuals belonging to low-income who receive subsidies. Non-PBI members consist of persons working in formal sectors with stable income, self-employed individuals, informal workers, farmers, and others who do not benefit from subsidies [28].

This research integrated six covariates at the district level, including GDP per capita (measured in USD), population density (people/km^2), and coverage per health facility (in km^2). Data on GDP per capita, population density, and the total number of health facilities were obtained from the Indonesian Statistics Bureau [30]. The average coverage of health facility was calculated by dividing the size of district by the sum of primary healthcare facilities and hospitals. Three classifications were established based on the coverage extent of a single health facility: less than or equal to 50 km2 ($\leq$ 50 km2), 50 to 200 km2, and greater than or equal to 200 km2 ($\geq$200 km2).

2.4. Statistical analysis

A series of line charts were created to depict the trends of climatic variables, alongside the trend of pregnancy complications from January 2015 to December 2022. In each chart, the vertical axes represented temperature in degree Celsius, precipitation in millimetres, relative humidity as a percentage, solar radiation in W/m^2, and the proportion of pregnancy complications as a percentage, respectively. The horizontal axis denotes the timeline in months.

Pregnancy cases were nested within individuals (level 1), districts (level 2) and provinces (level 3) as the clusters of data to analyse the associations between exposures and outcome using multilevel Poisson regression. The analysis included assessing the relative risk (RR) and 95% confidence interval (CI) to ascertain the relationships. We conducted a sequence of models. Model 1 included climatic factors and individual characteristics. Model 2 included Model 1 with district characteristics. The analysis employed R version 4.4.0 with lme4 package [31]. To evaluate the robustness of the findings, a sensitivity analysis of Model 2 was conducted by restricting the sample to urban areas, where geographic boundaries are relatively smaller and population density is higher. This restriction was intended to reduce potential exposure misclassification arising from spatial aggregation.

An ethical clearance exemption for this study was granted by the National Research and Innovation Agency (Badan Riset dan Inovasi Nasional) in Indonesia.

3. Results

Based on Fig. 1, the average monthly temperature (A) fluctuated between 23 °C and 28 °C, with peaks consistently between January and May, often exceeding 26 °C. Lower temperatures were typically observed around May and September, dipping slightly below 25.5 °C. For precipitation (B), a cyclical pattern was evident, reflecting Indonesia's wet and dry seasons. The highest rainfall, often over 250 mm, occurred from December to February, with another notable increase from October to December. The driest months were July and August, with precipitation often below 100 mm. Solar radiation (C) remained relatively stable, averaging between 350 and 520 W/m^2. Peaks, usually above 450 W/m^2, were noted from July to September, while the lowest levels, frequently below 350 W/m^2, occurred in November and December. Humidity (D) showed a consistent cyclical pattern, with peaks often exceeding 85% early in the year and around May, and lower levels in September.

Fig. 2 illustrates the monthly trend in the proportion of pregnancy complications in Indonesia from January 2015 to December 2022. The figure delineates a downward trend in the proportion of pregnancy complications over the observed period, with a particularly marked decline from 2020 onwards. Initial fluctuations in the proportion of complications were observed, with significant peaks and troughs until around March 2016. Following around March 2016, a gradual decline appeared, although the trend remained relatively unstable, exhibiting periodic fluctuations of increases and decreases until mid-2020. From mid-2020 onwards, the trend demonstrated a more pronounced decline, becoming particularly evident in early 2021, after which it stabilized with minor fluctuations. The proportion consistently remained around 2% for the remainder of the period until December 2022.

Table 1 provides a description of pregnancies in the sample from 2015 to 2022. Overall pregnancies happened among those aged around 29 years old, slightly higher that pregnancy with complications (31.2) and consistent across those with long treatment duration and high expenses. Regarding subsidy status, 49.5% of pregnancies with complications were covered by JKN subsidy. District characteristics highlight the mean GDP per capita at 3.98 (per 1000 USD) with a standard deviation of 4.58, and the population density at 22.91 per 100 km^2. Health facility coverage varied, with 56.69% of pregnancies in districts with coverage less than 50 km^2, and the majority of pregnancies occurred in rural areas (72.21%). For those with long treatment duration, 60.47% were in districts with less than 50 km^2 coverage and 75.72% resided in rural areas. Pregnancies with high expenses also predominantly occurred in non-subsidized individuals (60.50%).

Table 2 presents the results of a multilevel Poisson regression model analysing the incidence of pregnancy complications, duration, and

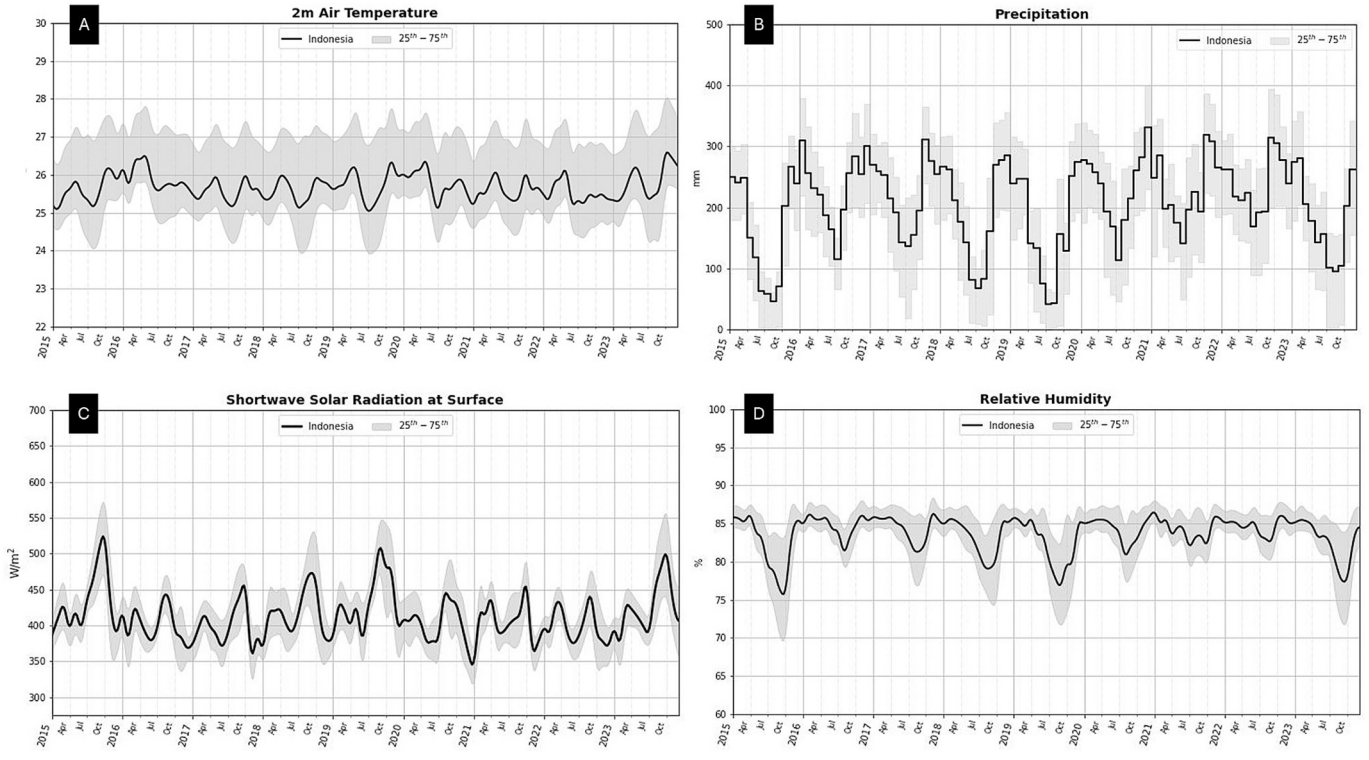


Fig. 1. Monthly average trend of temperature (A), precipitation (B), solar radiation at surface (C), and relative humidity (D) in Indonesia from 2015 to 2022.

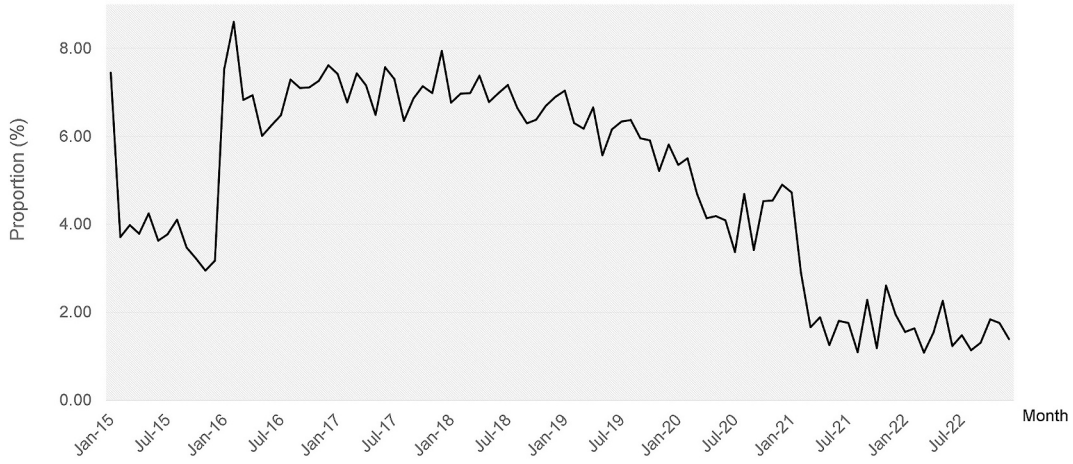


Fig. 2. Trend of pregnancy complication proportion from 2015 to 2022 in Indonesia.

Table 1
Description of pregnancies in the sample from 2015 to 2022, in percentages unless otherwise specified.

	Total of pregnancy	Pregnancy complication		
		Total	Long treatment duration	High expense
n	*342,432*	*22,136*	*4421*	*11,062*
Individual characteristics				
Age (per 10 years)*	2.90 (0.58)	3.12 (0.66)	3.13 (0.67)	3.15 (0.64)
Subsidy status				
Subsidized	42.28	49.5	49.02	39.50
Non-subsidized	57.72	50.5	50.98	60.50
District characteristics				
GDP per-capita (per 1000USD)*	3.98 (4.58)	3.98 (4.85)	3.84 (4.86)	4.26 (5.07)
Population density (per 100km^2)*	22.91 (40.75)	23.22 (40.65)	20.42 (38.91)	27.44 (44.53)
Health Facility coverage				
$<$ 50 km^2	56.69	60.47	53.92	61.64
50–200 km^2	26.04	23.22	26.33	22.93
$\geq$ 200 km^2	17.27	16.31	19.75	15.42
Type of residence				
Rural	72.21	75.72	77.11	70.68
Urban	27.79	24.28	22.89	29.32

medical expense related to pregnancy complications from January 2015 to December 2022. In Model 1, the analysis revealed that for every 5 °C increase in temperature, there was a decrease in pregnancy complication incidence, with an RR of 0.86 (95%CI:0.83–0.89). Similarly, an increase in precipitation (per 100 mm) was associated with a lower risk of pregnancy complications (RR = 0.97, 95%CI:0.95–0.98). Conversely, higher solar radiation (per 50 W/m^2) and humidity (per 10 g/m^3) were associated to increased risks, with relative risks of 1.02 (95% CI:1.01–1.04) and 1.12 (95%CI:1.05–1.19), respectively. However, the risk of longer treatment and higher medical expenses due to pregnancy complications were not statistically significant with temperature, solar radiation, precipitation, and relative humidity. After adjusting with district characteristics in Model 2, the magnitude and direction of risk remained.

The risks of pregnancy complication and associated treatment duration and medical expense by individual and district characteristics were also presented in Table 2. By individual characteristics, the risk of pregnancy complications rose significantly with a 10-year increase in age (RR:1.84; 95%CI:1.80–1.88) and were lower among individuals who received subsidy compared to who did not receive (RR: 0.70; 95% CI:0.68–0.72). The risk of higher associated medical expense was higher among older individuals (RR:1.06; 95%CI:1.03–1.09) and among those who did not received subsidy compared to who received (RR:1.47; 95% CI:1.41–1.53). After adjusting with district characteristics in Model 2, the magnitude and direction of risk remained. By district characteristics, the risk of pregnancy complication was negatively associated with urban (RR:0.85; 95%CI:0.74–0.97) compared to rural areas. Furthermore, the risk of having associated higher medical expenses was positively associated with an increase in population density (RR:1.02; 95% CI:1.01–1.03). The sensitivity analysis results presented in Supplementary Table 1 show only minor changes in the magnitude of the associations for all outcomes, while the direction and statistical significance remained consistent.

4. Discussion

4.1. Potential limitations

There are several limitations that should be considered when interpreting the findings of this study. First, the JKN data sample registry lacked detailed information on individual health behaviours and key clinical confounders, such as BMI, parity, chronic hypertension, diabetes, smoking status, and prenatal care utilization. The absence of these variables may introduce residual confounding, as these factors are known to influence the risk of pregnancy complications. Second, the JKN data sample primarily captured information related to claims. Therefore, women with pregnancy complications who were not enrolled in or did not utilize JKN may not be captured, potentially leading to underestimation of the true disease burden. Third, climate exposure was measured at the district level and assigned uniformly to individuals, which may not reflect true individual exposure and could introduce ecological bias. In addition, trimester-specific exposure and gestational

Table 2
Results of multilevel Poisson regression model with the pregnancy complication status, treatment duration, and medical expense related to pregnancy complication as the dependent variable, from January 2015 to December 2022.

	Pregnancy complication incidence *n = 342,432*)			Long treatment duration (*n = 22,136*)			High medical expense (*n = 22,136*)		
	%	Relative Risk (95% CI)		%	Relative Risk (95% CI)		%	Relative Risk (95% CI)	
		Model 1	Model 2		Model 1	Model 2		Model 1	Model 2
Temperature (per 5C)*	4.87 (0.50)	0.86 (0.83–0.90)	0.86 (0.83–0.89)	4.92 (0.48)	1.00 (0.98–1.03)	1.01 (0.93–1.10)	4.87 (0.49)	0.97 (0.92–1.02)	0.97 (0.92–1.02)
Solar radiation (per 50 W/m^2)*	8.00 (0.95)	1.02 (1.01–1.04)	1.02 (1.01–1.04)	8.01 (0.97)	0.99 (0.98–1.01)	0.99 (0.95–1.04)	8.01 (0.94)	1.01 (0.98–1.04)	1.01 (0.98–1.04)
Precipitation (per 100 mm)*	2.19 (1.19)	0.97 (0.95–0.98)	0.97 (0.95–0.98)	2.16 (1.19)	1.01 (0.99–1.02)	1.01 (0.97–1.04)	2.20 (1.18)	1.00 (0.98–1.02)	1.00 (0.98–1.02)
Humidity (per 10%)*	8.50 (0.31)	1.12 (1.05–1.19)	1.12 (1.05–1.19)	8.49 (0.32)	0.99 (0.95–1.03)	1.00 (0.87–1.16)	8.49 (0.31)	0.96 (0.88–1.06)	0.97 (0.88–1.06)
Individual characteristics									
Age (per 10 years)*	3.12 (0.66)	1.84 (1.80–1.88)	1.84 (1.80–1.88)	3.13 (0.67)	1.00 (0.99–1.02)	1.02 (0.97–1.06)	3.15 (0.64)	1.06 (1.03–1.09)	1.06 (1.03–1.09)
Subsidy status									
Subsidized	7.57	Ref	Ref	24.51	Ref	Ref	39.87	Ref	Ref
Non-subsidized	5.66	0.70 (0.68–0.72)	0.70 (0.68–0.72)	24.09	1.01 (0.99–1.03)	1.03 (0.97–1.09)	59.88	1.47 (1.41–1.53)	1.46 (1.41–1.52)
District characteristics									
GDP per-capita (per 1000USD)*	3.98 (4.85)		1.00 (0.99–1.01)	3.84 (4.86)		1.00 (0.99–1.01)	4.26 (5.07)		1.00 (0.99–1.01)
Population density (per 100km^2)*	23.22 (40.65)		0.99 (0.98–1.01)	20.42 (38.91)		1.00 (0.99–1.01)	27.44 (44.53)		1.02 (1.01–1.03)
Health Facility coverage									
$<$ 50 km^2	6.9		Ref	21.85		Ref	50.94		Ref
50–200 km^2	5.77		1.06 (0.95–1.18)	27.17		1.05 (0.92–1.21)	49.36		1.04 (0.94–1.15)
$\geq$ 200 km^2	6.1		1.10 (0.96–1.25)	29.06		1.14 (0.96–1.35)	47.26		0.96 (0.86–1.08)
Type of residence									
Rural	6.78		Ref	24.99		Ref	46.65		Ref
Urban	5.65		0.85 (0.74–0.97)	22.21		1.06 (0.89–1.26)	60.35		1.09 (0.97–1.23)

age alignment could not be assessed due to data limitations. Fourth, outcome identification relied on ICD-10 codes from administrative data, which may be subject to coding inaccuracies and variability across healthcare facilities. Although diagnoses recorded in secondary care facilities are generally more reliable, misclassification cannot be entirely ruled out. Fifth, seasonal variation was not explicitly accounted for in the analysis, which may influence both climate exposure and pregnancy outcomes. Finally, although some associations were statistically significant, effect sizes were small and should be interpreted with caution. The proposed mechanisms are based on existing literature and remain biologically plausible rather than causal.

4.2. Interpretation of results

This study found that low temperatures increased the risk of pregnancy complications. This finding aligns with the systematic review by Hlimi (2015) but is inconsistent with a study by Qu et al. (2021) [15], reporting the associated risk of heatwaves on pregnancy complications in the United States [20]. In temperate areas, heatwaves can cause extreme hot temperatures, leading to severe health and environmental consequences. Unlike regions with four distinct seasons, tropical regions experience relatively stable narrow ranges of temperatures throughout the year. Particularly, Indonesia's tropical climate results in a consistently narrow temperature range, with moderation provided by oceanic and maritime influences. Low temperatures can lead to vasoconstriction [4], which can increase stress on the cardiovascular system and the risk of eclampsia and preeclampsia. Lee et al. (2013) identified that individuals from tropical regions exhibit greater susceptibility to cold-induced vasoconstriction and cold-related injuries when exposed to cooler climates [32]. However, they appear to have developed physiological adaptations to higher temperatures, enhancing their tolerance to heat.

Our findings highlighted that the increased risk of pregnancy complications was associated with periods of low precipitation levels. This is inconsistent with a study from South Africa by Immink et al. (2008) [16] but aligns with a study conducted in Thailand, reporting an increased risk during the dry season [17]. Geographically, Thailand and Indonesia are situated at similar latitudes and within the same regional climate zone, which may subject them to comparable climate conditions. An increased precipitation indicated by frequent rains can help to dilute and wash away air pollutants [33]. Without rain, contaminants can potentially accumulate in the atmosphere, leading to prolonged exposure to harmful substances among pregnant women. Furthermore, when precipitation is low, humans adapt to this condition by producing more vasopressin—a hormone helps to maintain the body's fluid balance—to conserve body fluid [34]. An increase in vasopressin causes increased retention of salt and water which increases blood pressure and may increase the risks of hypertensive disorder.

Higher solar radiation was associated with an increased risk of developing eclampsia and preeclampsia. Ivanov et al. (2018) stated that excessive exposure to such radiation may seriously contribute to increase in oxidative stress [35]. Moreover, excessive solar radiation can disrupt the balance of vitamin D in the human body. For instance, a study by Agarwal et al. (2023) found an excessive vitamin D intake is linked to vascular calcification due to the deposition of calcium in the walls of blood vessels [36]. Furthermore, high solar radiation can

influence behaviour and environmental conditions, such as increased indoor air pollution due to more frequent use of air conditioning and reduced physical activity [37].

An elevated risk of eclampsia and preeclampsia was associated with higher humidity levels. This aligns the findings of a study conducted in Zimbabwe by Wacker et al. (1998) [18]. When the air is saturated with moisture, the evaporation process slows down or stops, making it difficult for the body to regulate its temperature [8]. As a result, the body compensates by producing more sweat to cool down, leading to excessive sweating and discomfort. This disruption may result in excessive fluid intake to compensate, leading to hyponatremia (low sodium levels). Hyponatremia can directly impact neural function, making the brain more vulnerable to seizures, a key characteristic of eclampsia [38,39].

We found weak evidence that climatological factors affected the severity or eclampsia and preeclampsia, indicated by the longer duration and higher expense of the associated medical treatment. Marthias et al. (2022) found that the access of maternal health services in Indonesia were significantly improved after the expansion of JKN [40], potentially enabling quicker diagnosis and treatment to prevent such severity. In South Korea, a significant reduction in the incidence of preeclampsia is reported following the implementation of a universal voucher scheme for pregnant women [41]. By enabling broader access to regular prenatal care, this voucher system facilitated increased maternal health check-ups, which, in turn, contributed to the early identification and prevention of complications such as preeclampsia. Therefore, pregnant women may receive timely and appropriate treatment without incurring significant financial burdens [40].

5. Conclusions

This study presents evidence of the effects of climatic factors on the incidence of eclampsia and preeclampsia, which align with studies undertaken in areas with comparable latitudinal and meteorological characteristics. This result also highlights the necessity of incorporating localized climate-sensitive early warning systems into public health frameworks, particularly in regions susceptible to extreme weather variations. A coordinated approach involving multiple stakeholders is essential, with policymakers prioritizing the integration of climate variables into maternal health programs, ensuring robust policy implementation, and equipping healthcare providers to address climate-related health challenges. Such efforts are crucial for building a climate-resilient health system that can protect pregnant women, reduce maternal mortality, and enhance maternal health outcomes.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.preghy.2026.101484.

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